05-MOTORS-SERVOS-DRIVERS-AND-RELAYS

Item #	Item Description	Technical Specifications & Model Details	Required Quantity	Asset Classification
01	**TB6612FNG Dual Motor Driver Module for Arduino/Other Microcontroller**	**High-Efficiency Dual MOSFET H-Bridge Driver Module**; 1.2A continuous / 3.2A peak per channel; ultra-low internal resistance; runs cool without heatsink; operating logic 2.7V–5.5V; motor voltage 4.5V–13.5V; PWM speed control *(Replaces outdated L293D / L298)*	**10 Units**	**Consumable**
02	**SP Electron L293D Motor Driver Shield For Microcontrollers**	4-Channel driver shield using bipolar pulse-width modulation; plugs directly onto Arduino Uno / Mega headers; drives up to 4 DC motors or 2 stepper motors with screw terminal power connections	**5 Units**	**Consumable**
03	**L293D Dual H-Bridge Motor Driver Module**	Standalone BJT-based dual H-bridge motor driver board; controls speed and direction of 2 DC motors or 1 stepper motor; up to 600mA per channel; operating voltage 4.5V–12V; legacy demonstration driver	**2 Units**	**Consumable**
04	**L298P High-Power Motor Shield**	Arduino Uno form-factor shield featuring L298 dual full-bridge BJT driver; drives up to 2 DC motors (2A per channel); heavy screw terminals for external motor power; onboard buzzer and servo header interface	**2 Units**	**Consumable**
05	**PCA9685 16-Channel 12-bit PWM / Servo Driver**	16-channel independent PWM controller with I2C bus; driven by only 2 microcontroller pins (SDA, SCL); 12-bit resolution; handles 16 servos	**3 Units**	**Consumable**
06	**ULN2003 Stepper Motor Driver Board with 28BYJ-48 Stepper**	5V unipolar 4-phase stepper motor paired with ULN2003 Darlington driver board with 4-phase status indicator LEDs	**10 Sets**	**Consumable**
07	**1-Channel 5V/3.3V Relay Module with Optocoupler & Trigger Selector**	5V DC relay module with PC817 optocoupler isolation, transistor driver, status LED, and **High/Low level trigger selection jumper**; 10A 250V AC	**6 Units**	**Consumable**
08	**2-Channel 5V/3.3V Relay Module with Optocoupler & Trigger Selector**	Dual 5V DC relay board with independent optoisolated inputs, status LEDs, High/Low trigger selection jumpers, and NO/COM/NC terminals	**15 Units**	**Consumable**
09	**4 Channel 5V Relay Module with Optocoupler Isolation**	4-Channel 5V relay module with optocoupler isolation (10A 250V AC / 30V DC), Songle SRD-05VDC-SL-C with individual status LEDs for UNO, Raspberry Pi, ESP32, robotics & home automation	**5 Units**	**Consumable**
10	**Robocraze 8 Channel Relay Module 5V with Optocoupler Isolation**	Arduino Compatible PLC Control Board; 8 independent 10A high-current relays with optocoupler isolation, status indicator LEDs, and screw terminal blocks for smart home automation, IoT, robotics & industrial control	**3 Units**	**Consumable**
11	**150 RPM Plastic BO Gear Motor with Wheel**	3V–9V DC dual-shaft straight gearbox motor (150 RPM output speed @ 6V) complete with matching 65mm rubber-tread wheel	**30 Pairs**	**Consumable**
12	**150 RPM BO Motor (Standalone)**	3V–9V DC battery-operated single-shaft plastic gear motor (150 RPM)	**10 Units**	**Consumable**
13	**12V High-Torque DC Geared Motor**	12V DC heavy-duty metal-gear motor; 150 RPM output; 6mm D-shaped center output shaft	**10 Units**	**Consumable**
14	**Submersible Mini Water Pump**	12V DC, 3W submersible centrifugal brushless water pump with outlet nozzle; max head 1.5 metres	**3 Units**	**Consumable**
15	**Micro Positional Servo — Plastic Gear (SG90)**	9g micro positional servo motor; 0° - 180° pulse angle rotation; operating voltage 4.8V - 6.0V; stall torque 1.8 kg-cm; horns included	**30 Units**	**Consumable**
16	**Metal-Gear Micro Servo (MG90S)**	High-durability micro servo with brass and aluminum gears; 0° - 180° rotation; stall torque 2.2 kg-cm	**10 Units**	**Consumable**

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17	**Standard High-Torque Positional Servo**	Full-size metal-gear positional servo (MG995 / MG996R); torque 10–12 kg-cm; double ball bearings; 0° - 180° control	**5 Units**	**Consumable**
18	**Continuous Rotation Metal Servo (360°)**	Full-size continuous rotation servo (360° speed/direction control); high-torque metal gear train	**5 Units**	**Consumable**
19	**3-Channel Servo & ESC Tester (CCPM)**	Standalone digital servo tester with 3 modes (Manual speed dial, Neutral auto-center, Automatic window sweep); tests up to 3 servos simultaneously	**3 Units**	**Consumable**